

The Effect of Strabismus and Anisometropia on the Development of Amblyopia in Synaptic Plasticity Models

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1 Intro

2 Methods

Amblyopia is the most common cause of vision loss in children, caused by refractive errors (anisometropic) or misalignment of the eyes (strabismus)¹.

What about Section sec. 3?

Our model does not include a direct mechanism for binocular rivalry, but has the competition between the eyes driven by the temporal competition between patterns as is typical of the BCM model^{citeBCMpapers?}.

The neuron develops normal, oriented, binocular receptive fields in this balanced environment as shown in Figure 2.

3 Results

4 References

1. de Zárata, B.R., and Tejedor, J. (2007). Current concepts in the management of amblyopia. Clinical ophthalmology (Auckland, NZ) 1, 403.



Figure 1: Original images

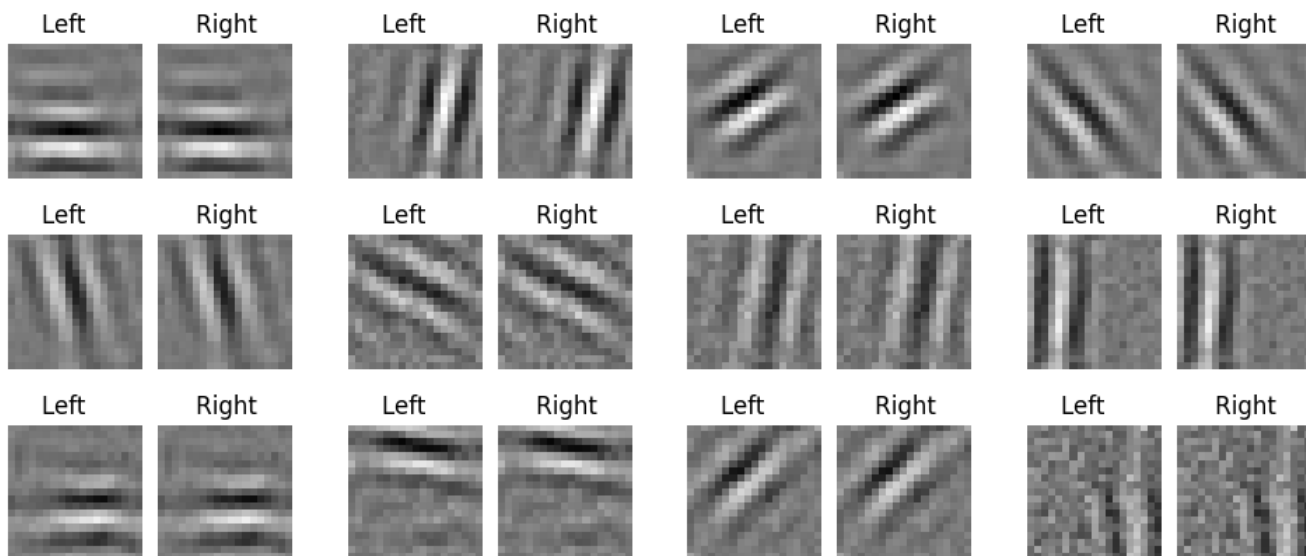


Figure 2: Receptive fields formed in a normal, balanced, natural image environment. Every simulation produces oriented, binocular receptive fields. Shown are the synaptic weights where white denotes strong weights and black denotes weak weights.